CLIMATESCOPE: CLIMATE RISK ALERT SYSTEM

Mini Project Report

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Abstract

The ClimateScope application is a data-driven climate monitoring and alert system developed using Python and Streamlit. The system analyzes global weather data and provides insights into temperature, humidity, wind speed, and air quality.

A key feature of the system is the Climate Risk Alert System, which evaluates environmental conditions and categorizes risk levels into low, moderate, and high. The application includes interactive dashboards, country-wise comparisons, and visualization tools.

The project aims to help users understand climate conditions and make informed decisions based on real-time data insights.

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1 Introduction

Climate change and environmental conditions are becoming increasingly important in today's world. Monitoring climate parameters such as temperature, humidity, and air quality helps in understanding environmental risks.

The ClimateScope application is designed to provide a comprehensive dashboard for analyzing climate data. It integrates visualization, data filtering, and risk prediction into a single platform.

2 Objectives

- To analyze global climate data
- To build an interactive dashboard using Streamlit
- To develop a Climate Risk Alert System
- To provide country-wise climate insights
- To implement data visualization techniques
- To ensure system reliability through testing

3 Technologies Used

- Python
- Streamlit
- Pandas
- Plotly
- PyCountry
- Visual Studio Code

4 System Architecture

User $\downarrow$ Login $\downarrow$ Dashboard $\downarrow$ Data Processing $\downarrow$ Visualization $\downarrow$ Risk Analysis $\downarrow$ Output

5 Implementation

The application was developed using Streamlit and consists of multiple modules:

- Login System: Secure authentication using hashed passwords
- Dashboard: Displays global climate summary
- Overview Module: Shows trends and insights
- Comparison Module: Compares climate between countries
- Volatility Module: Measures data variability
- Map Module: Visualizes global data using maps
- Profile Module: Displays country-specific metrics
- Risk Module: Calculates climate risk score

The system uses real-world climate data and processes it dynamically based on user inputs.

6 Climate Risk Alert System

The risk system calculates a score based on temperature, humidity, and air quality.

$$\textbf{Risk Score} = (\textbf{Temperature} \times \textbf{0.4}) + (\textbf{Humidity} \times \textbf{0.2}) + (\textbf{PM2.5} \times \textbf{0.4})$$

Based on the score:

- Low Risk
- Moderate Risk
- High Risk

The system also provides alerts and recommendations to users.

7 Testing and Validation

The application was tested using functional testing methods.

A separate Python file (test_cases.py) was created to validate:

- Risk calculations
- Data filtering
- Comparison module
- Volatility calculations
- Map data
- Profile metrics
- Edge cases such as empty data and zero values

All test cases passed successfully, ensuring system reliability.

8 Results

The application successfully provides:

- Interactive dashboards
- Climate trend analysis
- Country comparisons
- Risk predictions
- Visual representations of data

The system is efficient and user-friendly.

9 Conclusion

The ClimateScope application demonstrates how data analytics and visualization can be used to understand climate conditions. The Climate Risk Alert System adds value by providing meaningful insights and alerts.

The project successfully meets all objectives and provides a scalable solution for climate monitoring.

10 Future Enhancements

- Integration with real-time APIs
- Mobile application development
- AI-based prediction models
- User-specific notifications
- Advanced forecasting techniques

11 References

- Python Documentation
- Streamlit Documentation
- Pandas Library
- Plotly Library